Joint-Embedding Predictive Architectures and Latent Residual Conditioning for Multivariate Time-Series Anomaly Detection

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***Abstract*—Deep reconstructive detectors for multivariate time-series anomaly detection minimize observation-space error, fitting stochastic sensor noise that produces heavy-tailed residuals and severe false-alarm flooding under automated threshold calibration. We investigate how Joint-Embedding Predictive Architectures (JEPAs) condition predictive residuals in a low-dimensional latent space under distribution shift, examining unconstrained models, Operator-Entropy JEPA (OpEntropy JEPA) with spectral conditioning, and Reynolds-Stress JEPA with Crouzeix-contractive dynamics and positive semidefinite stress closure. Across 33 continuous telemetry streams from seven physical datasets under chronological held-out Streaming Peaks-Over-Threshold calibration, our primary finding is relative false-alarm suppression rather than universal detection superiority or exact attainment of the 0.1% nominal design risk. Latent predictive scoring reduces held-out false-positive rates three-fold under retrospective buffered attribution (6.7% to 10.9% for base latent predictors versus 30.0% to 30.7% for TimesNet and TranAD) and eight- to ten-fold under strictly causal streaming (1.9% to 2.8% versus 22.2% to 24.3%), while point F1 and precision-recall ranking remain comparable to reconstructive baselines. Controlled ablations isolate two distinct mechanisms: layer-normalization bottlenecks compress residual kurtosis, whereas learned latent projections halve calibration-to-holdout distribution drift relative to normalized observation forecasters. On a fixed Koopman backbone, equivalence testing confirms that spectral entropy and barrier terms preserve full operator rank without altering point F1 within a 0.01 to 0.03 margin, while the OpEntropy transition architecture shortens detection delay relative to an unconstrained predictor. Static thresholding still overshoots its 0.1% design target by 15- to 64-fold out of sample, driven primarily by inter-block distribution shift and serial dependence.**

***Index Terms*—Time-series anomaly detection, Multivariate analysis, Threshold calibration, Extreme value theory, Representation learning.**

I. Introduction

Autonomous anomaly detection in multi-sensor telemetry monitors operational health across aerospace systems, Earth observation, cyber-physical infrastructure, and wearable digital healthcare [1], [2]. In complex deployments such as NASA SMAP and MSL rovers, industrial process loops, and biometric sensor arrays, dozens of channels concurrently track power buses, thermal gradients, vibrations, and kinematics [3]. Operational monitors must identify subsystem faults without overwhelming operators with nuisance alerts.

False-Alarm Flooding Under Automated Threshold Calibration: Operational monitoring requires automated decision rules that convert continuous anomaly scores into binary alarms using nominal calibration data alone. Prevailing deep detectors (TimesNet, TranAD, Anomaly Transformer) minimize point-level reconstruction error in raw observation space [4]–[6]. Because raw telemetry contains ambient sensor jitter and channel-scale disparities, observation-space residuals exhibit heavy-tailed excess kurtosis ($\gamma_2 > 8$) and strong serial clustering. When paired with automated extreme-value thresholding such as Streaming Peaks-Over-Threshold (SPOT) [7] at nominal design risk $q = 10^{-3}$ (0.1%), these heavy-tailed residuals suffer severe calibration breakdown on unseen nominal data, triggering false alarms across 22%–31% of nominal timesteps.

Latent Residual Conditioning and Scope of Claims: A detector that predicts the *summary* of a future window, rather than every raw sample, ignores sensor jitter that no model can forecast. We therefore study Joint-Embedding Predictive Architectures (JEPAs) [8], which forecast a compact 32-dimensional latent summary of the next 64 steps from the preceding 256, without synthetic anomaly injection [9]. Beyond unconstrained multi-layer perceptron predictors (`TS-JEPA`), we formulate physically grounded and structurally constrained variants: an *Operator-Entropy JEPA* (OpEntropy JEPA) regularized by Koopman spectral entropy and a Bregman energy barrier, and a *Reynolds-Stress JEPA* enforcing Crouzeix numerical-range contractivity and positive semidefinite second-moment closure. We treat these structures as geometric conditioning priors on latent trajectories, reporting them as honest regularizer controls without inflated claims of aggregate detection superiority. The evaluation asks three separate questions: (i) how much do latent residuals reduce false alarms under automated calibration, and does calibration reach its nominal 0.1% risk target; (ii) does detection quality hold across operating points and fault classes; and (iii) how long does an operator wait between a fault and an alarm.

A. Contributions

Each contribution below states one finding in words. The supporting numbers sit in the cited tables and figures.

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TABLE I
KEY OPERATIONAL AND STATISTICAL TERMS USED THROUGHOUT THE MANUSCRIPT.

Term	Meaning
Latent residual	Gap between the predicted and observed 32-dimensional summary of the next window.
SPOT	Streaming extreme-value rule that sets an alarm threshold from nominal scores alone.
False-positive rate (FPR)	Fraction of nominal timesteps that raise an alarm.
Event recall	Fraction of true anomaly segments with at least one alarm.
Buffered	Retrospective scoring that spreads each window score over its 64 steps.
Causal	Scoring that uses only past data and assigns each score to its final step.
Calibration drift	Shift between residuals on calibration data and on later nominal data, measured by the Kolmogorov–Smirnov distance.
Excess kurtosis	Tail heaviness of the residual distribution; large values mean rare, extreme residuals.
Equivalence test	Test that shows two models differ by less than a stated margin (Section V).

1) **Why latent scoring lowers false alarms:** Across 14 controlled backbone arms, a layer-normalization (LayerNorm) bottleneck bounds heavy residual tails, and a learned latent projection also keeps held-out residuals close to calibration residuals (Section VI-E, Table II(e)).
2) **Fewer false alarms, comparable detection:** On 33 streams from seven datasets, latent scoring cuts held-out false alarms several-fold relative to reconstruction baselines under both retrospective (buffered) and strictly causal evaluation, and it recovers more faults at a fixed low false-alarm budget. Point-wise F1 (Point-F1) and precision-recall area under the curve (PR-AUC) stay comparable to the baselines, and quiet single-channel faults remain weakly detected (Tables II(e) and III, Fig. 3(b)).
3) **Physics-inspired regularizers condition but do not drive detection:** Proposition 1 proves non-degeneracy of the Koopman transition operator, while algebraic Crouzeix bounds guarantee contractive dynamics. Equivalence tests show that removing operator or stress terms leaves Point-F1 unchanged within a narrow ± 0.010 margin, while structured prediction shortens detection delay and suppresses causal false alarms (Tables III and II).
4) **A measured calibration gap:** Standard extreme-value thresholding still exceeds its 0.1% design risk out of sample. A refitted permutation test attributes most of the excess to distribution shift between calibration and test blocks, and a smaller share to serial dependence (Table IV).

Roadmap. Section II reviews related work and Section III describes the benchmark streams. Section IV defines the architecture and scoring rule, and Section V defines the evaluation protocol and statistics. Sections VI and VII report ablations and benchmark results, and Section VIII concludes.

II. RELATED WORK

Observation-Space Forecasting and Reconstructive Detectors: Deep time-series anomaly detectors predominantly operate in raw observation space by training a network to forecast future sensor readings or reconstruct input windows, flagging timesteps where the point-level observation error $\|\mathbf{X} - \hat{\mathbf{X}}\|_F^2$ exceeds a threshold. Autoregressive long short-term memory (LSTM) networks with non-parametric dynamic thresholding [3] model sequential dependencies directly on raw spacecraft telemetry channels. More recent Transformer and spectral reconstructors introduce specialized attention and frequency modules: TranAD [6] combines adversarial two-phase training with self-conditioning sub-series attention; Anomaly Transformer [4] contrasts prior Gaussian kernel associations against series-level self-attention to amplify association discrepancies; and TimesNet [5] folds 1D sequences into 2D tensors along dominant Fast Fourier Transform (FFT) periods so 2D convolutional kernels can capture intra- and inter-period variations.

Despite strong ranking performance on clean benchmarks, observation-space reconstructors share a structural vulnerability under automated thresholding on noisy multi-sensor arrays. Because point-level losses penalize reconstruction error across every channel equally, the network is forced to model high-frequency stochastic noise, quantization jitter, and background environmental fluctuations. In high-dimensional streams ($K \geq 25$), aggregating raw reconstruction errors across channels accumulates heavy-tailed residual spikes, referred to statistically as *leptokurtic* tails (excess kurtosis $\gamma_2 > 0$), which destabilize extreme-value threshold fitting and inflate false alarms whenever baseline operating conditions shift [10].

Contrastive and Joint-Embedding Predictive Architectures: To avoid reconstructing raw sensor noise, self-supervised representation methods map windows into a compact latent embedding space. Contrastive detectors construct positive and negative pairs: TS2Vec [11] maximizes contextual agreement across hierarchical sub-series; Neural Contextual Anomaly Detection (NCAD) [9] contrasts full window embeddings against context-only embeddings using injected synthetic anomalies; and DCdetector [12] employs dual patch-wise and in-patch attention branches without a reconstruction loss. However, contrastive classifiers depend on whether synthetic perturbations or patch discrepancies match test-time fault morphology, yielding lower calibrated point detection on diverse continuous telemetry (0.058 and 0.070 macro Point-F1 for NCAD and DCdetector vs. 0.160 for latent predictive scoring across 11 datasets, Table III(d)).

Joint-Embedding Predictive Architectures (JEPAs) [8], [13], [14] eliminate both raw observation decoding and synthetic negative pairs. Rather than reconstructing inputs, a JEPA encodes a historical context window into a latent vector z_{ctx} and predicts the target representation z_{tgt} of a future window produced by a separate target encoder. Because trivial constant embeddings would zero out latent prediction error, non-contrastive architectures prevent representation collapse either through architectural asymmetry (updating the target encoder via an exponential moving average of online weights [15]) or through explicit variance-covariance regularization such as VICReg [16], Barlow Twins [17], and batch whitening [18]. Prior time-series representation studies evaluate embeddings via post-hoc threshold sweeps; how latent predictive residuals behave under strictly unsupervised extreme-value threshold calibration remains unexplored.

Koopman Operator Theory and Physics-Inspired Priors: Multi-sensor telemetry reflects the underlying trajectories of physical dynamical systems [19]. Neural ordinary differential equations (ODEs) [20] parameterize continuous velocity fields in observation space, whereas Physics-Informed Neural Networks (PINNs) [21] and Hamiltonian or Lagrangian networks

[22], [23] enforce partial differential conservation laws directly on physical state coordinates. However, rigid differential constraints require known governing equations and suffer from stiff competing gradient pathologies on uncalibrated sensor arrays [24].

Koopman operator theory [25], [26] provides a softer dynamical prior: nonlinear dynamics can be lifted into an observable embedding space where time evolution is governed by a linear operator. Data-driven Koopman networks [27]–[29] learn neural embeddings jointly with a finite-dimensional transition matrix $\mathbf{K}$. Here we regularize the singular value spectrum of $\mathbf{K}$ via Von Neumann spectral entropy and a smooth Bregman energy barrier, maintaining a strict distinction between *physics-inspired* algebraic priors (which shape abstract learned latent coordinates) and *physics-informed* conservation laws, and testing whether operator conditioning yields downstream detection gains or statistical equivalence relative to an unconstrained predictor.

Extreme-Value Threshold Calibration and Evaluation Rigor: Unsupervised streaming deployment requires converting continuous scores into binary alarms using nominal calibration data alone. Streaming Peaks-Over-Threshold (SPOT) [7] fits a Generalized Pareto Distribution (GPD) to score exceedances above a high calibration quantile u to extrapolate a threshold at design risk q. However, classical extreme-value theory assumes independent and identically distributed (i.i.d.) samples. Continuous telemetry violates this assumption through short-range serial correlation, which causes threshold exceedances to occur in multi-step bursts quantified by the Ferro–Segers extremal index [30], and through non-stationary regime drift between calibration and test blocks. Concurrently, recent benchmark audits [10], [31], [32] have shown that "point-adjusted" F1 (which credits an entire anomaly segment if a single timestep triggers an alarm) severely inflates scores; we therefore evaluate all models using unadjusted Point-F1, PR-AUC, causal vs. buffered attribution, and matched realized false-alarm operating curves.

III. Multi-Sensor Internet-of-Things and Remote Sensing Benchmark Suite

Our benchmark suite comprises 11 datasets spanning 45 continuous telemetry streams sourced from NASA spacecraft archives [3], the University of California, Irvine (UCI) Machine Learning Repository [33], [34], TimeEval [10], and mTSBench [35]: 33 streams are evaluated under the primary multi-seed held-out calibration protocol and 12 reserved for operational case studies. Streams span four physical domains: *spacecraft telemetry* (SMAP and MSL entities, with two short-nominal MSL streams reserved; SWAN-SF solar magnetograms [36]); *wearable kinematics* (six Daphnet Freezing-of-Gait (FoG) streams [33]: 64 Hz tri-axial accelerometry for Freezing-of-Gait events, so S=64 is a 1-s horizon); *industrial processes* (14 Gasoil Heating Loop (GHL) streams, the Genesis pick-and-place testbed, and GECCO water-distribution chemistry/hydrology at 1-minute resolution); and *smart infrastructure* (Room-Occupancy environmental sensing [37], six CICIDS edge network-flow streams, CalIt2 people-flow [34], and Metro I-94 highway traffic).

Each stream provides nominal training sequences and labeled anomaly test sequences; channels are standardized by training statistics with $\boldsymbol{\sigma}$ floored at 10^{-2} against flatline phases. We evaluate the standardized subsets studied in the literature [3], [10] (of 55 SMAP and 27 MSL archive entities) for strict matching and reproducible runtime.

To eliminate circular in-sample threshold tuning, each nominal sequence is split chronologically into training (65%), nominal calibration (17.5%), and held-out nominal evaluation (17.5%) blocks separated by an unobserved 64-step isolation gap (Section V-A). Stream inclusion in the primary multi-seed protocol required: (i) $N_{\text{cal}} \geq 500$ calibration windows after splitting, and (ii) tractable stride-1 multi-seed runtime. Exactly 33 streams across seven datasets satisfy both criteria (6 Daphnet, K=9; 14 GHL, K=16; 4 MSL, K=55; 6 SMAP, K=25; Genesis, K=18; Metro, K=5; SWAN, K=38). The remaining 12 streams (CalIt2, K=2; two Room-Occupancy, K=5; two short-nominal MSL; GECCO, K=9; six 72-channel CICIDS flows) appear in operational incident case studies and a secondary 11-dataset, 130-stream single-seed baseline cross-check (Table III(d)); the dyadic front-end ablation (Table II(b)) includes SMD `machine-1-2` (K=38).

Two external-validity caveats bound interpretation: (i) SMAP and MSL contain documented mislabeled segments and point anomalies [32], so per-dataset scores represent upper bounds on operational detectability; and (ii) because streams within SMAP, MSL, Daphnet, and GHL share hardware and environmental characteristics, streams within a dataset are not independent domains (N=7 dataset clusters). We therefore report unweighted Dataset-Macro means, between-stream and within-stream seed standard deviations (SDs), leave-one-dataset-out (LODO) sensitivity, and dataset-clustered permutation tests (Table III(c)), restricting conclusions strictly to the evaluated entities and sampling regimes.

IV. Methodology: Operator-Entropy and Reynolds-Stress Joint-Embedding Predictive Architectures

A. Problem Formulation and Latent Space Prediction

Consider a continuous multivariate telemetry stream $\mathbf{X} \in \mathbb{R}^{T \times K}$ recording K sensor channels over T timesteps. As illustrated in Fig. 1 (Stage 1), a sliding window of total length $W = C + S = 320$ steps advances across the stream at stride 1 and partitions the local series at split index t into two contiguous blocks:

1) **Historical context window ($\mathbf{x}_{\text{ctx}} = \mathbf{X}[t-C : t, :] \in \mathbb{R}^{C \times K}$):** The preceding $C = 256$ observations leading up to index t, which establish the current nominal operating regime, phase, and baseline amplitude of the system.
2) **Suspect target horizon ($\mathbf{x}_{\text{tgt}} = \mathbf{X}[t : t+S, :] \in \mathbb{R}^{S \times K}$):** The subsequent $S = 64$ observations spanning from t to the online decision timestamp $\tau = t + S - 1$, which are inspected to determine whether the system has departed from its expected trajectory.

Unlike reconstructive detectors that minimize pointwise observation error $\|\mathbf{X} - \hat{\mathbf{X}}\|_F^2$ and inherit leptokurtic sensor noise (Sections I and II), we shift prediction into a compact

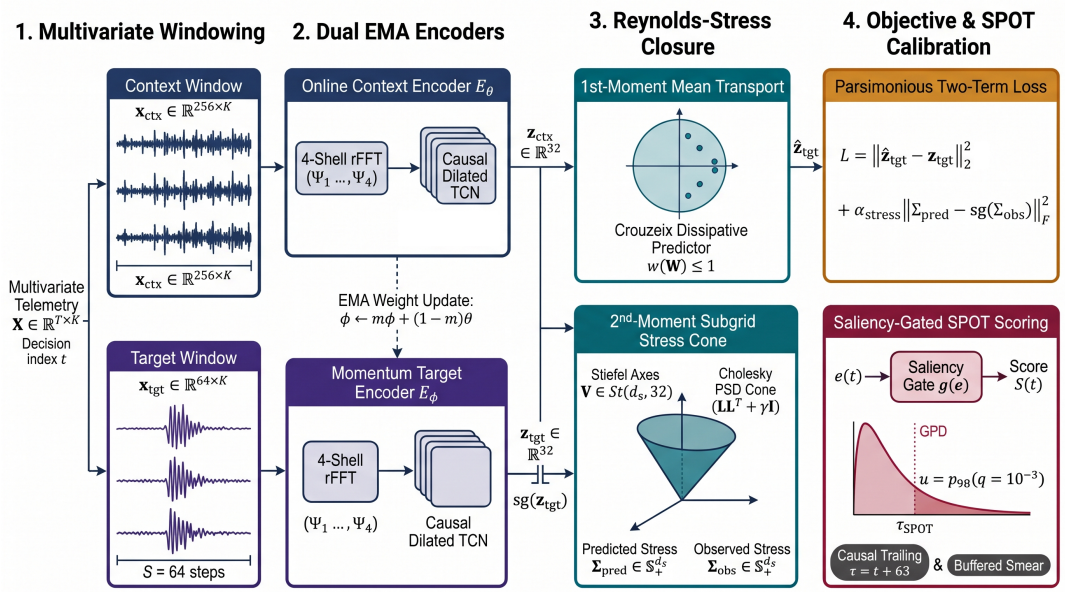


Fig. 1. Four-stage `Reynolds-Stress JEPA` architecture. (1) Multiscale windowing splits telemetry at index t into context ($\mathbf{x}_{\text{ctx}} \in \mathbb{R}^{256\times K}$) and target ($\mathbf{x}_{\text{tgt}} \in \mathbb{R}^{64\times K}$) blocks processed by a 4-shell dyadic real fast Fourier transform (rFFT) front-end and causal dilated temporal convolutional network (TCN). (2) Dual online (E_θ) and exponential moving average (EMA) target (E_ϕ) encoders map windows into $\mathbb{R}^{32}$. (3) Reynolds-stress closure forecasts first-moment transport ($\hat{\mathbf{z}}_{\text{tgt}}$) via a Crouzeix dissipative predictor ($w(\mathbf{W}) \le 1$) and second-moment subgrid covariance ($\mathbf{\Sigma}_{\text{pred}}, \mathbf{\Sigma}_{\text{obs}} \in \mathcal{S}_+^{d_s}$) on a Stiefel frame $\mathbf{V} \in \text{St}(d_s, 32)$. (4) Parsimonious two-term training objective and saliency-gated SPOT calibration under causal trailing ($\tau = t + 63$) and buffered smear mappings.

32-dimensional latent space ($D = 32$, Fig. 1) using a Joint-Embedding Predictive Architecture (JEPA) [8]. An online context encoder E_θ compresses $\mathbf{x}_{\text{ctx}}$ into a context vector $z_{\text{ctx}} \in \mathbb{R}^D$, while a separate target encoder E_ϕ compresses $\mathbf{x}_{\text{tgt}}$ into a target vector $z_{\text{tgt}} \in \mathbb{R}^D$. Rather than training E_ϕ by direct gradient descent (which would allow both encoders to map all inputs to a constant vector), the target weights ϕ are updated as an exponential moving average (EMA) of the online weights θ, $\phi_k \leftarrow m_k\phi_{k-1} + (1-m_k)\theta_k$ with a cosine momentum schedule $m_k \in [0.996, 0.9995]$ [13], [15]. By predicting the summary embedding z_{tgt} from z_{ctx}, the model discards uninformative channel-level jitter while preserving structural trajectory departures.

B. Latent Predictor Architectures: Operator-Entropy and Reynolds-Stress JEPA

We evaluate three nested latent predictor architectures sharing the same multiscale encoder backbone and coordinate saliency gate: an unconstrained multi-layer perceptron predictor (`TS-JEPA`), a spectral-regularized Koopman transition (`Operator-Entropy JEPA`), and a structurally constrained second-moment closure architecture (`Reynolds-Stress JEPA`).

1) Operator-Entropy JEPA (Spectral Conditioning): **Formulation.** Koopman operator theory [25], [26] represents nonlinear dynamics as a linear map on learned observables, $\hat{z}_{\text{tgt}} = \mathbf{K}z_{\text{ctx}}$ with $\mathbf{K} \in \mathbb{R}^{D\times D}$. Trained on prediction error alone, $\mathbf{K}$ can concentrate energy into a few directions (rank collapse) or shrink toward zero (norm collapse). We regularize both the shape and scale of the singular value spectrum $\sigma_1 \ge \cdots \ge \sigma_D \ge 0$ of $\mathbf{K}$ via $\mathcal{R}_{\text{spectral}}(\mathbf{K}) = \Delta\mathcal{H}(\mathbf{K}) + \beta_{\text{norm}}\Phi_{\text{energy}}(\mathbf{K})$. Normalizing squared singular values $\lambda_i = \sigma_i^2$ into $p_i = (\lambda_i + \epsilon_{\log})/\sum_{j=1}^{D}(\lambda_j + \epsilon_{\log})$ ($\epsilon_{\log} = 10^{-12}$) defines the Von Neumann spectral entropy $\mathcal{H}_{\text{spectral}}(\mathbf{K}) = -\sum_{i=1}^{D} p_i \log p_i$ and the *Spectral Entropy Deficiency* relative to uniform equipartition $\mathbf{u} = [1/D, \ldots, 1/D]^\top$:

$$\Delta\mathcal{H}(\mathbf{K}) = \log D - \mathcal{H}_{\text{spectral}}(\mathbf{K}) = \mathcal{D}_{\text{KL}}(\mathbf{p} \,\|\, \mathbf{u}) \ge 0. \quad (1)$$

Because $\Delta\mathcal{H}(c\mathbf{K}) = \Delta\mathcal{H}(\mathbf{K})$ is scale-invariant for $c > 0$, a convex Bregman barrier on the energy ratio $r(\mathbf{K}) = \|\mathbf{K}\|_F^2/D$ prevents norm decay without hard gradient clipping ($\epsilon = 10^{-6}$):

$$\Phi_{\text{energy}}(\mathbf{K}) = r(\mathbf{K}) - 1 - \log\big(r(\mathbf{K}) + \epsilon\big) + \log(1+\epsilon) \ge 0. \quad (2)$$

Combining both terms yields the Operator-Entropy training objective:

$$\mathcal{L}_{\text{Op-Entropy}} = \frac{1}{BD}\sum_{b=1}^{B}\big\|\mathbf{K}z_{\text{ctx}}^{(b)} - z_{\text{tgt}}^{(b)}\big\|_2^2 + \lambda_{\text{ent}}\mathcal{R}_{\text{spectral}}(\mathbf{K}) + \beta_{\text{Lyap}}\mathcal{L}_{\text{Lyap}}, \quad (3)$$

where B is the batch size, $\beta_{\text{norm}} = 1.0$, $\lambda_{\text{ent}} = 0.5$, and $\beta_{\text{Lyap}} = 0.02$ aligns the trace-normalized batch covariance of $\mathbf{K}\,\text{sg}(z_{\text{ctx}})$ to z_{tgt} using stop-gradient $\text{sg}(\cdot)$. Because $\epsilon = 10^{-6}$ smooths the logarithm in (2), we verify that $\mathbf{K} = \mathbf{0}$ cannot become a false equilibrium during training.

Proposition 1 (Operator Non-Degeneracy at the Origin)**.** *Let* $\mathbf{M} = \frac{1}{BD}\sum_{b=1}^{B} z_{tgt}^{(b)} z_{ctx}^{(b)\top}$ *denote the context-to-target cross-covariance under (3) with* $\lambda_{ent}, \beta_{norm} > 0$*: (i)* Non-zero gradient for finite $\epsilon > 0$: *At* $\mathbf{K} = \mathbf{0}$, $\nabla_{\mathbf{K}}\mathcal{R}_{spectral}(\mathbf{0}) = \mathbf{0}$*; whenever* $\mathbf{M} \neq \mathbf{0}$, $\nabla_{\mathbf{K}}\mathcal{L}_{Op\text{-}Entropy}(\mathbf{0}) = -2\mathbf{M} \neq \mathbf{0}$*, so* $\mathbf{K} = \mathbf{0}$ *is never a stationary point. (ii)* Exact barrier limit: *As* $\epsilon \to 0^+$, $\lim_{\|\mathbf{K}\|_F \to 0} \Phi_{energy}(\mathbf{K}) = +\infty$*, forming an infinite barrier against norm collapse.*

Proof. At $\mathbf{K} = \mathbf{0}$, uniform weights $p_i = 1/D$ minimize $\Delta\mathcal{H}$ ($\nabla_{\mathbf{K}}\Delta\mathcal{H}(\mathbf{0}) = \mathbf{0}$), and $\nabla_{\mathbf{K}}\Phi_{\text{energy}}(\mathbf{0}) = \mathbf{0}$ by the chain rule on (2). Differentiating the prediction term at $\mathbf{K} = \mathbf{0}$ leaves

$-2\mathbf{M} \neq \mathbf{0}$, while $\epsilon \to 0^+$ as $r(\mathbf{K}) \to 0$ drives $-\log r(\mathbf{K}) \to +\infty$. □

Remark (Scope of Regularization): Proposition 1 prevents collapse of the transition matrix $\mathbf{K}$, while encoder representation diversity is maintained by the EMA target encoder and stop-gradient decoupling rather than extra variance penalties such as VICReg [16] or Barlow Twins [17] (Table II(c)).

2) Reynolds-Stress JEPA (Second-Moment Subgrid Closure): **Physical Motivation and Structural Parsimony.** Predicting only the window mean $\hat{z}_{\text{tgt}}$ (first moment) can miss variance-shifting faults whose mean stays near nominal. Inspired by Reynolds averaging in turbulence, which splits a flow into its resolved mean $\bar{z}$ and subgrid covariance $\mathbf{\Sigma}$, Reynolds-Stress JEPA predicts both moments while enforcing stability and positive semidefinite (PSD) geometry directly inside the network layers rather than via extra loss penalties:

1) *Stable mean prediction (Crouzeix-contractive layers):* Each predictor layer splits its weight $\mathbf{W}$ into skew-symmetric rotation $\mathbf{A} = \frac{1}{2}(\mathbf{W} - \mathbf{W}^\top)$ and spectrally normalized symmetric damping $\mathbf{S} = \frac{1}{2}(\mathbf{W} + \mathbf{W}^\top)$, forming $\mathbf{W}_{\text{eff}} = \tau\mathbf{A} + \sigma_{\text{sym}}\mathbf{S}/(\|\mathbf{S}\|_2 + \epsilon)$. This keeps the numerical range inside the unit stability disk without eigenvalue clipping, while lag-8 Hankel and 8-shell radial Fourier filters damp high-frequency context jitter.
2) *Subgrid covariance and PSD closure (Stiefel projection):* Alongside the pooled summary $\bar{z}_{\text{tgt}} \in \mathbb{R}^D$ (mean, max, terminal, standard deviation, log-energy, and first-difference statistics), the encoder centers the 64-channel TCN sequence $h(\tau) - \bar{h}$ and projects it via an orthonormal basis $\mathbf{Q}_{\text{St}} \in \text{St}(D, 64)$ into zero-mean fluctuations $z'_{\text{tgt}}(\tau) \in \mathbb{R}^D$. A second orthonormal frame $\mathbf{V} \in \text{St}(d_s, D)$ ($d_s = 8$, via QR factorization) compresses $z'_{\text{tgt}}(\tau)$ into the observed stress matrix $\mathbf{\Sigma}_{\text{obs}} \in \mathcal{S}_+^{d_s}$, while a midpoint closure head on $\bar{z} = \frac{1}{2}(z_{\text{ctx}} + \hat{z}_{\text{tgt}})$ predicts $\mathbf{\Sigma}_{\text{pred}} = \mathbf{L}\mathbf{L}^\top + \gamma\mathbf{I}_{d_s} \in \mathcal{S}_+^{d_s}$ under a two-term objective:

$$\begin{aligned}\mathcal{L}_{\text{Reynolds}} = &\frac{1}{BD}\sum_{b=1}^{B}\left\|\hat{z}_{\text{tgt}}^{(b)} - z_{\text{tgt}}^{(b)}\right\|_2^2 \\ &+ \frac{\alpha_{\text{stress}}}{Bd_s^2}\sum_{b=1}^{B}\left\|\mathbf{\Sigma}_{\text{pred}}^{(b)} - \text{sg}\big(\mathbf{\Sigma}_{\text{obs}}^{(b)}\big)\right\|_F^2,\end{aligned} \tag{4}$$

with $\alpha_{\text{stress}} = 0.1$ and no auxiliary loss penalties.

C. Anomaly Scoring and Operational Detection Latency

At test time, sliding windows are processed at stride 1. Let $\mathbf{e}(t) = z_{\text{tgt}}(t) - \hat{z}_{\text{tgt}}(t) \in \mathbb{R}^D$ denote the latent prediction residual, and let $g(\mathbf{e})$ denote the homogeneous coordinate saliency gate in (7). Unconstrained `TS-JEPA` scores windows via $\mathcal{S}(t) = g(\mathbf{e}(t))$. Because the final LayerNorm layer projects latent vectors onto the sphere of radius $\sqrt{D}$ and strips amplitude, `Operator-Entropy JEPA` tracks pre-LayerNorm radial energy $(e_{\text{ctx}}, e_{\text{tgt}})$ with $e_{\text{pred}} = e_{\text{ctx}}\|\hat{z}_{\text{tgt}}\|_2/(\|z_{\text{ctx}}\|_2 + 10^{-6})$, while `Reynolds-Stress JEPA` combines radial leakage with 4-octave Littlewood–Paley Parseval shell-energy leakage $\Delta E_{\text{LP}}(t)$ and saliency-gated subgrid stress closure discrepancy:

$$\mathcal{S}_{\text{OpEnt}}(t) = g\big(\mathbf{e}(t)\big) + 0.2\,\frac{|e_{\text{tgt}}(t) - e_{\text{pred}}(t)|}{e_{\text{tgt}}(t) + e_{\text{pred}}(t) + 10^{-6}}, \tag{5}$$

$$\begin{aligned}\mathcal{S}_{\text{Rey}}(t) = &\, g\big(\mathbf{e}(t)\big) + 0.2\,\Delta E_{\text{LP}}(t) \\ &+ \frac{0.2}{\sqrt{k_s}}\, g\big(\text{vec}(\mathbf{\Sigma}_{\text{obs}} - \mathbf{\Sigma}_{\text{pred}})\big),\end{aligned} \tag{6}$$

where $k_s = \min(K, d_s)$ denotes the active subgrid dimension.

Three-Part Operational Causality Framework: To avoid confounding data access, window buffering, and alarm timing, we formalize online evaluation under three criteria (illustrated in Fig. 2):

(a) **Data causality ($\mathcal{F}_\tau$-measurability):** At decision timestamp $\tau = t + S - 1$, the detector observes only historical data $\mathbf{X}[t-C:\tau, :]$; neither encoder accesses observations beyond τ.

(b) **Score-emission latency ($S - 1 = 63$ steps) vs. fault-detection delay (Δt):** Because z_{tgt} summarizes the S-step target block $\mathbf{X}[t : t + S, :]$, window score s_t is emitted at the trailing edge $\tau = t + S - 1$ (Fig. 2, Row 3). This creates a structural score-emission latency of $S-1 = 63$ steps (0.98 s on 64 Hz Daphnet, 10.5 min on 0.1 Hz SMAP/MSL/GHL, 63 min on GECCO, and 63 h on hourly Metro). This structural lag is distinct from empirical fault-detection delay $\Delta t = t_{\text{alarm}} - t^*$, which measures the elapsed steps from ground-truth anomaly onset t^* to the first timestamp $t_{\text{alarm}} \geq t^*$ whose attributed score exceeds threshold τ_{SPOT}. Shortening the horizon to $S \in \{1, 4\}$ eliminates score-emission latency when rapid tripwire response is required (Section VII-D).

(c) **Strictly causal trailing vs. retrospective buffered attribution:** Under *strictly causal trailing attribution*, window score s_t is assigned only to the decision timestamp $\tau = t + S - 1$ for zero-lookahead streaming. Under *retrospective buffered smear attribution*, s_t is averaged across all steps in $[t + 1, t + S]$, which introduces fixed-lag smoothing and can smear post-onset scores up to S steps before t^*. Both regimes are reported separately throughout (Table III(a),(b)).

D. Shared Architectural Components and Thresholding

1) Homogeneous Coordinate Saliency Gating: In high-dimensional streams ($K \geq 25$), faults affecting a single subsystem can be diluted by nominal variance across the remaining coordinates. To amplify sparse latent deviations while preserving exact degree-1 homogeneity $\mathcal{D}_{\text{sal}}(c\mathbf{e}) = c\,\mathcal{D}_{\text{sal}}(\mathbf{e})$ for any $c > 0$ without heuristic box clamping, we normalize coordinate magnitudes by the root-mean-square scale $\sigma_{\mathbf{e}} = \sqrt{\frac{1}{D}\sum_{j=1}^{D} e_j^2 + \epsilon}$ before applying the softmax coordinate saliency gate $g(\mathbf{e}) \equiv \mathcal{D}_{\text{sal}}(\mathbf{e})$ to $\mathbf{e} = \hat{z}_{\text{tgt}} - z_{\text{tgt}} \in \mathbb{R}^D$:

$$\begin{aligned}\mathcal{D}_{\text{sal}}(\mathbf{e}) &= (1-\alpha)\|\mathbf{e}\|_2 + \alpha\sqrt{D\sum_{d=1}^{D} w_d e_d^2}, \\ w_d &= \frac{\exp\big(|e_d|/(\tau_g\sigma_{\mathbf{e}})\big)}{\sum_{j=1}^{D}\exp\big(|e_j|/(\tau_g\sigma_{\mathbf{e}})\big)},\end{aligned} \tag{7}$$

with fixed blend $\alpha = 0.5$ and temperature $\tau_g = 0.5$ across all latent models. Input channels are encoded with uniform weights in the deployed pipeline; an auxiliary context-variance input-channel gate is evaluated in Section VI-C.

2) Multiscale Frequency Front-End: For multivariate streams ($K > 1$), a causal depthwise 1D convolutional stem filters each sensor channel independently prior to 1×1 cross-channel mixing ($K \leq 1$ bypasses the stem via an explicit identity short-circuit). Before the dilated temporal convolutional network (TCN) backbone, each projected channel is split across $J = 4$ smooth dyadic frequency shells on a 256-point real Fast Fourier Transform (rFFT) grid normalized into an exact amplitude partition of unity ($\sum_{j=1}^{J} \Psi_j[\omega_m] \equiv 1$ to within 1.2×10^{-7}). Each shell is processed by a causal 1D convolution with kernel sizes $\{15, 9, 5, 3\}$ from coarse to fine scales and recombined via a softmax scale router conditioned on the causal right-edge state, channel root-mean-square amplitude, and standardized octave log-energy. Because smooth Fourier shells overlap at band edges, this module acts as a multiscale frequency front-end rather than a strictly orthogonal wavelet bank (Table II(b)).

3) Extreme-Value Threshold Calibration (SPOT): To convert continuous scores $\mathcal{S}(t)$ into binary alarms $\hat{y}_t = \mathbb{I}(\mathcal{S}(t) > \tau_{\text{SPOT}})$ using nominal calibration data alone, we apply Streaming Peaks-Over-Threshold (SPOT) [7]. Rather than assuming the entire score distribution is Gaussian, SPOT isolates the upper tail above the 98th calibration percentile $u = p_{98}$ and fits the exceedances $Y = \mathcal{S}(t) - u > 0$ to a Generalized Pareto Distribution (GPD) with tail-heaviness shape parameter ξ and scale parameter σ via Grimshaw's maximum-likelihood algorithm [38]. Inverting the fitted GPD tail at target false-alarm probability $q = 10^{-3}$ (one expected false alarm per 1,000 nominal steps) yields the closed-form threshold:

$$\tau_{\text{SPOT}} = u + \frac{\sigma}{\xi}\left[\left(\frac{qN_{\text{cal}}}{N_u}\right)^{-\xi} - 1\right], \tag{8}$$

where N_{cal} is the number of nominal calibration windows and N_u is the number of exceedances above u. All primary benchmarks use standard unadjusted SPOT (τ_{SPOT}). In Section VII-F, we also audit two heuristic kurtosis adjustments: total-threshold scaling $\tau_{\text{eff}} = \tau_{\text{SPOT}}[1 + \frac{1}{6}\tanh(\gamma_2)]$ and excess-only scaling $\tau_{\text{exc}} = u + (\tau_{\text{SPOT}} - u)[1 + \frac{1}{6}\tanh(\gamma_2)] \geq u$.

V. Experimental Protocol and Evaluation Metrics

A. Experimental Setup and Hyperparameters

All architectures use an identical streaming pipeline with context window $C = 256$ and target horizon $S = 64$ ($W = C + S = 320$), trained at stride 16 and evaluated at stride 1. Context and target representations ($D = 32$) are extracted by the multiscale frequency front-end followed by a 6-block causal dilated TCN (64 filters, kernel size 5, dilations $\{1, 2, 4, 8, 16, 32\}$). Multi-statistic temporal pooling (terminal, mean, max, standard deviation, Log-Ricci energy, and first-order total variation) projects features through a linear bottleneck before LayerNorm and dropout ($p = 0.20$). Models are trained for 20 epochs on nominal sequences using AdamW (learning rate $10^{-3} \rightarrow 10^{-5}$ via cosine annealing, weight decay 10^{-4}, batch size 32).

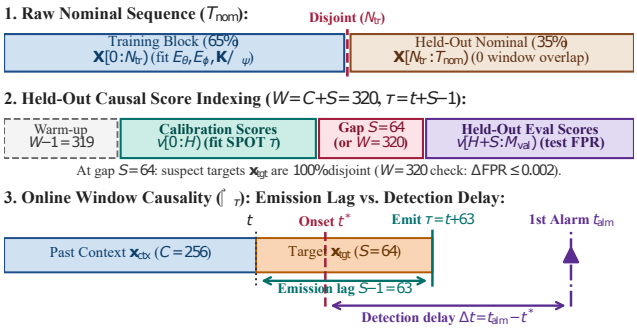


Fig. 2. Chronological split, boundary window indexing, and online causality protocol (C=256, S=64, W=320). Row 1 separates raw training (65%) and held-out nominal (35%) observations with zero window overlap across N_{tr}. Row 2 indexes causal held-out scores $v[0 : M_{\text{val}})$ into context warm-up ($W-1$=319), calibration $v[0 : H)$, isolation gap S=64 (or full-window W=320), and held-out evaluation $v[H+S : M_{\text{val}})$. Row 3 distinguishes structural score-emission lag ($S-1$=63) from empirical fault-detection delay ($\Delta t = t_{\text{alm}} - t^*$).

a) Held-Out Calibration Protocol and Boundary Isolation: To evaluate out-of-sample threshold calibration without circularity, each stream's raw nominal sequence $\mathbf{X}_{\text{nom}} \in \mathbb{R}^{T_{\text{nom}} \times K}$ is partitioned chronologically (Fig. 2, Row 1) into a *training block* $\mathbf{X}_{\text{tr}} = \mathbf{X}_{\text{nom}}[0 : N_{\text{tr}})$ ($N_{\text{tr}} = \lfloor 0.65\, T_{\text{nom}} \rfloor$) and a disjoint *held-out nominal block* $\mathbf{X}_{\text{val}} = \mathbf{X}_{\text{nom}}[N_{\text{tr}} : T_{\text{nom}})$. Standardization parameters and neural weights are fit on $\mathbf{X}_{\text{tr}}$ windows alone, guaranteeing zero raw-observation or window overlap across N_{tr}.

Within $\mathbf{X}_{\text{val}}$ ($N_{\text{val}} = T_{\text{nom}} - N_{\text{tr}}$), sliding windows of length $W = 320$ are constructed strictly inside $\mathbf{X}_{\text{val}}$ (Fig. 2, Row 2). Under causal trailing attribution, the first $W-1 = 319$ observations serve as context warm-up, yielding $M_{\text{val}} = N_{\text{val}} - W + 1$ valid score timestamps $v[0 : M_{\text{val}})$. Splitting v at midpoint $H = \lfloor M_{\text{val}}/2 \rfloor$ defines the *nominal calibration block* $v_{\text{cal}} = v[0 : H)$ (used to fit the SPOT threshold at $u = p_{98}, q = 10^{-3}$) and the *held-out evaluation block* $v_{\text{hold}} = v[H + S : M_{\text{val}})$, excluding $S = 64$ intermediate windows $v[H : H + S)$. Because suspect target windows have length $S = 64$, the last calibration target and first evaluation target are 100% disjoint; the first $C = 256$ evaluation steps share passive historical context with calibration observations, mirroring online streaming where past nominal history forms context for current predictions. A stricter full-window isolation check ($G = W = 320$, excluding $v[H : H + 320)$ so even passive context is disjoint across the seven streams with $M_{\text{val}}/2 > 420$) leaves stream-macro causal FPR virtually unchanged (2.07% $\rightarrow$ 2.18% for OpEntropy and 3.02% $\rightarrow$ 3.24% for TS-JEPA versus 12.15% $\rightarrow$ 13.97% for TimesNet and 9.05% $\rightarrow$ 9.73% for TranAD). All primary tables use fixed standard SPOT thresholds during testing.

b) Multi-Seed Replication, Non-Parametric Hypothesis Testing, and Equivalence Bounds: Principal models on the 19 non-GHL streams are evaluated across five random seeds ($\{42, 123, 456, 7, 2024\}$); the 14 GHL streams ($> 10^6$ observations each) and ablation controls use three seeds. Because random seeds on the same stream are not independent observations, all metrics are averaged across seeds within each stream first, treating the telemetry stream ($n = 33$ or $n = 19$) as the independent statistical unit. To compare models without

assuming Gaussian error distributions across diverse domains, we employ three statistical tools:

1) **Paired Wilcoxon signed-rank test and false-discovery-rate adjustment:** The Wilcoxon signed-rank test is a non-parametric paired hypothesis test that evaluates whether the median difference between two models across the same streams is zero by ranking the absolute magnitudes of their per-stream score differences. Unlike a paired Student's t-test, it prevents a single extreme outlier stream from skewing the comparison. When testing multiple model pairs simultaneously, we adjust p-values using the Benjamini–Hochberg false-discovery-rate (FDR) procedure ($q_{\text{FDR}} = 0.05$) to cap the expected proportion of false statistical positives, and report 95% stream-resampled bootstrap confidence intervals (CIs; $2{,}000$ resamples).
2) **Two One-Sided Tests (TOST) for statistical equivalence:** In a standard two-sided test, an insignificant result ($p > 0.05$) indicates only inconclusive evidence rather than equal performance. To test whether a physics-inspired regularizer produces practically negligible change relative to an unregularized control, TOST reverses the null hypothesis by testing whether the true mean difference $\mu_1 - \mu_2$ lies outside a prespecified equivalence margin $[-\Delta, +\Delta]$ via two one-sided tests: $H_{01} : \mu_1 - \mu_2 \leq -\Delta$ (p_{lower}) and $H_{02} : \mu_1 - \mu_2 \geq +\Delta$ (p_{upper}). Formal equivalence is established if $p_{\text{TOST}} = \max(p_{\text{lower}}, p_{\text{upper}}) < 0.05$ (meaning the 90% paired confidence interval falls strictly inside $[-\Delta, +\Delta]$). We prespecify $\Delta = \pm 0.03$ Point-F1 for fixed-backbone regularizer controls ($\approx 13\%$ of baseline F1, $< 0.09\times$ between-stream standard deviation) and $\Delta = \pm 0.05$ for cross-architecture comparisons, and report narrow-margin sensitivity checks down to $\Delta \in \{\pm 0.010, \pm 0.015, \pm 0.020\}$ (Table II(a)).
3) **Distributional tail and drift diagnostics (γ_2 and D_{KS}):** To quantify why anomaly scores trigger false alarms, we measure two summary statistics on nominal data: (i) *excess kurtosis* $\gamma_2 = \mathbb{E}[(s-\mu)^4]/\sigma^4 - 3$, which measures tail heaviness relative to a Gaussian distribution ($\gamma_2 = 0$), where *leptokurtic* scores ($\gamma_2 > 0$) exhibit heavy outlier-prone tails and *platykurtic* scores ($\gamma_2 < 0$) exhibit light, flat-topped tails compressed by normalization or temporal smoothing; and (ii) the two-sample *Kolmogorov–Smirnov (KS) drift statistic* $D_{\text{KS}} = \sup_x |F_{\text{cal}}(x) - F_{\text{hold}}(x)| \in [0, 1]$, which measures the maximum vertical gap between the empirical cumulative distribution functions of the calibration block (F_{cal}) and the held-out nominal evaluation block (F_{hold}), directly quantifying how severely the nominal baseline shifts over time.

B. Baseline Architectures, Fairness Checks, and Cohort Structure

We separate primary multi-seed conclusions (Cohorts 1–2 across 12 neural, linear, and non-parametric detectors) from secondary single-seed cross-checks. Two fairness audits verify that reconstructive baseline results under the unified 20-epoch, $W = 320$ pipeline are not artifacts of window length or implementation:

1) **Native published window lengths** (5 streams): Evaluating TranAD ($W = 10$) and TimesNet ($W = 96$) at their published window sizes raises Point-F1 (0.164 and 0.218), but observation residuals remain severely leptokurtic ($\gamma_2 = 448.3$ and 98.1 vs. 3.1 for TS-JEPA) with high nominal FPR (28.4% and 44.1%).
2) **Secondary 130-stream single-seed cross-check:** Reconstructors replicate their published point-adjusted F1 on SMD and MSL, and Table III(d) provides a secondary 130-stream, 11-dataset single-seed comparison against NCAD, DCdetector, and Anomaly Transformer.

C. Cohort Summary and Reproducibility Specifications

Experiments are organized into five non-overlapping cohorts:

- **Cohort 1 (33-stream primary multi-seed benchmark):** 19 non-GHL streams (5 seeds) and 14 GHL streams (3 seeds), reported in Tables III(a)–(c) and IV(a).
- **Cohort 2 (19-stream mechanism and operating-curve suite):** 14 controlled backbone arms across 5 seeds, reported in Table II(a),(e) and Fig. 3(b).
- **Cohort 3 (6-stream operator sweep and quiet-fault suite):** Factorial regularizer sweep (3 seeds) and six injected single-channel fault types, reported in Table II(c),(d).
- **Cohort 4 (5-stream horizon and frequency-shell panels):** Target-horizon latency sweep ($S \in \{1, 4, 16, 32, 64\}$, 3 seeds) and dyadic front-end ablation, reported in Fig. 3(a) and Table II(b).
- **Cohort 5 (130-stream single-seed baseline screen):** Secondary cross-check across 11 datasets against NCAD, DCdetector, and Anomaly Transformer (Table III(d)).

At $K = 25$ channels, model parameter counts are comparable across families: $285{,}701$ (OpEntropy JEPA), $289{,}061$ (TS-JEPA), $648{,}553$ (TimesNet), and $205{,}746$ (TranAD). Non-finite telemetry entries are zero-imputed with channel standard deviations floored at 10^{-2}; SPOT fits GPD parameters (ξ, σ) above $u = p_{98}$ via Grimshaw maximum-likelihood estimation (MLE) with a bounded quasi-Newton (L-BFGS-B) fallback ($\xi \in [-0.9, 10], \sigma \geq 10^{-6}$). All 195 primary multi-seed runs converged under the standard 20-epoch schedule on PyTorch 2.4 / CUDA 12.4 (NVIDIA RTX 5090, 0.60–0.84 ms per window, 184 MB of GPU memory); stream manifests, fitted thresholds, and source code are available at https://github.com/quaid281/NCAD.

D. Multi-Tier Statistical Evaluation Metrics

To guard against point-adjustment inflation, channel-averaging bias, and threshold gaming [10], [31], we evaluate models across three complementary tiers: (i) *primary calibration and accuracy endpoints*, comprising held-out nominal FPR, unadjusted Point-F1, and threshold-free PR-AUC; (ii) *operating characteristics*, evaluating event recall across deployable calibration quantiles (q^*, Fig. 3(b)) and across post-hoc matched realized test-FPR budgets ($f^* \in \{0.01, 0.05\}$, Table II(e)); and (iii) *secondary operational endpoints*, reporting unweighted Dataset-Macro Point-F1 (preventing the 14-stream GHL block from dominating), event recall (fraction of

contiguous anomaly segments with ≥ 1 alarm), mean detection delay Δt over detected events, and the constant-alarm baseline $\mathrm{F1}_{\text{all-pos}} = 2\pi/(1+\pi)$ at anomaly prevalence π.

VI. Ablation Studies and Analysis

A. Ablation 1: Operator Regularizer Control on Fixed Architecture

To test whether the spectral regularizer drives anomaly detection accuracy on its fixed Koopman backbone, Table II(a) compares Operator-Entropy JEPA against an unregularized control ($\lambda_{\text{ent}} = 0$) across all 19 non-GHL streams. Removing the spectral regularizer leaves buffered Point-F1 (0.233 vs. 0.235, paired difference -0.004, two-sided $p = 0.89$) and PR-AUC (0.351 vs. 0.365) virtually identical. Two One-Sided Tests confirm formal equivalence within the prespecified $\Delta = \pm 0.03$ margin ($p_{\text{TOST}} = 0.0007$) and across narrow-margin sensitivity checks down to $\Delta = \pm 0.020$ ($p_{\text{TOST}} = 0.0015$), $\Delta = \pm 0.015$ ($p_{\text{TOST}} = 0.0024$, with 90% CI $[-0.0112, +0.0029]$ strictly inside $[-\Delta, +\Delta]$), and $\Delta = \pm 0.010$ ($p_{\text{TOST}} = 0.0088$), as well as within ± 0.010 PR-AUC and ± 0.025 event recall.

This fixed-backbone regularizer control aligns with findings on the Reynolds-Stress JEPA architecture: isolating the base predictive residual from the quadratic stress closure discrepancy ($\alpha_{\text{stress}} = 0$) leaves buffered PR-AUC (0.281 vs. 0.285) and Point-F1 (0.167 vs. 0.148, paired difference -0.009, 90% CI $[-0.052, +0.032]$) comparable to the full Reynolds model while achieving 0.019 causal held-out FPR and leading matched-FPR event recall (0.457 at f^*=0.01 and 0.633 at f^*=0.05, Table II(a),(e)). Across both physics-inspired architectures, regularizer penalties condition latent representations without driving aggregate detection gains, confirming that false-alarm suppression is governed by the joint-embedding prediction framework rather than penalty additions.

B. Ablation 2: Factorial Isolation of the Entropy Term and Energy Barrier

Table II(c) crosses the spectral-entropy term and Bregman energy barrier across $\lambda_{\text{ent}} \in [0, 2.0]$ on six streams (three seeds). To quantify spectral diversity, we report the effective rank $\mathrm{RankMe} = \exp(-\sum_i p_i \log p_i) \in [1, 32]$ (which measures how many of the $D = 32$ dimensions carry active variance) and the operator condition number $\kappa = \sigma_{\max}/\sigma_{\min} \geq 1$ (the ratio of largest to smallest singular values, where $\kappa = 1$ denotes an ideal distance-preserving isometry). First, because the Koopman matrix $\mathbf{K}$ is initialized near the identity, the unregularized operator is already near full rank ($\mathrm{RankMe}(\mathbf{K}) = 31.53$ of 32, condition number $\kappa = 1.437$); increasing λ_{ent} tightens conditioning monotonically ($\kappa = 1.255$ at the deployed $\lambda_{\text{ent}} = 0.50$, and $\kappa = 1.154$ at $\lambda_{\text{ent}} = 2.00$) while lowering training loss variation ($\mathrm{CV}_{\mathcal{L}} : 0.565 \rightarrow 0.545$). Second, because $\mathcal{L}_{\text{Lyap}}$ applies a stop-gradient to z_{ctx}, encoder effective ranks remain invariant across all regularizer strengths ($\mathrm{RankMe}(z_{\text{ctx}}) \in [5.16, 5.54]$, $\mathrm{RankMe}(z_{\text{tgt}}) \in [11.54, 11.58]$). Third, detection and false-alarm rates remain stable across arms, with the deployed setting ($\lambda_{\text{ent}} = 0.50, \beta_{\text{norm}} = 1$) attaining the highest causal event recall (0.444 vs. 0.369 unregularized). On an eight-stream decoupling panel, pairing detached covariance alignment ($\beta_{\text{Lyap}} = 0.02$) with the smooth Bregman barrier cuts causal detection delay from 127.0 steps (TS-JEPA) and 80.2 steps ($\beta_{\text{Lyap}} = 0$) to 58.2 steps while raising causal Point-F1 from 0.044 to 0.094.

C. Ablation 3: Auxiliary Input-Channel Gating and Quiet-Fault Monitors

Comparing the auxiliary context-variance input-channel gate against uniform channel weighting on four streams confirms that uniform weighting achieves higher macro Point-F1 (0.118 vs. 0.077), so the deployed pipeline retains uniform input encoding. In Table II(d), injecting six single-channel fault types into six multivariate streams shows that quiet faults (flatlines, stuck sensors, dropouts) yield low point recall (0.02–0.07) across all latent gating variants because LayerNorm embeddings absorb constant single-channel offsets as nominal background. Auxiliary observation-space monitors, namely a rolling variance floor (`lowvar`) and maximum channel z-score (`dev_maxz`), recover quiet-fault recall (0.18–0.32) at the cost of inflating nominal FPR eight-fold (14.1%–14.9% vs. 1.9%).

D. Ablation 4: Multiscale Dyadic-Shell Front-End

In Table II(b), replacing the standard convolutional stem with the 4-shell Fourier front-end increases macro PR-AUC from 0.213 to 0.263 (+0.050, led by SMD and GECCO) and lowers calibration residual kurtosis ($1.409 \rightarrow 1.069$), while macro Point-F1 shifts from 0.118 to 0.078 due to conservative threshold tightening on SMAP `P-3`. The module thus acts as a ranking-conditioning stem rather than a uniform F1 gain.

E. Mechanism-Discrimination Study: Latent Space vs. Observation Space

Comparing the 14 controlled backbone arms across all 19 non-GHL streams in Table II(e) separates geometric score compression from representation drift reduction:

1) **Unnormalized observation forecasting (`TCN-ObsPred`):** Raw observation residuals exhibit heavy tail kurtosis ($\gamma_2 = 6.10$), severe calibration-to-holdout KS drift ($D_{\text{KS}} = 0.463$), and high causal false alarms (24.7%).
2) **LayerNorm observation bottleneck (`TCN-ObsPred-LN`):** Inserting a $D = 32$ LayerNorm bottleneck before decoding mechanically bounds excursions, compressing kurtosis to $\gamma_2 = 0.91$ and halving causal FPR to 11.7%, but leaves calibration-to-holdout drift elevated ($D_{\text{KS}} = 0.376$).
3) **Learned latent projection (`TS-JEPA`, `OpEntropy`, `Reynolds`):** Forecasting in learned latent coordinates halves calibration-to-holdout KS drift ($D_{\text{KS}} : 0.376 \rightarrow 0.193$–$0.274$, paired Wilcoxon $p \leq 0.001$) and reduces causal false alarms to 1.9%–8.6% (1.9%–2.8% without stress-closure scoring).

TABLE II

ABLATION AND MECHANISM-DISCRIMINATION STUDIES: (A) FIXED-BACKBONE REGULARIZER CONTROL AND TOST EQUIVALENCE (n=19, 5 SEEDS), (B) DYADIC FRONT-END (n=5), (C) OPERATOR REGULARIZER SWEEP (n=6, 3 SEEDS), (D) INJECTED QUIET-FAULT POINT RECALL (n=6), AND (E) 14-ARM BACKBONE COMPARISON (n=19, 5 SEEDS). ER IS EVENT RECALL AND MAD IS MEDIAN ABSOLUTE DEVIATION.

(a) Regularizer Control on Fixed Architecture & TOST Equivalence

Model / Control	Pt-F1	PR	FPR	ER	ΔF1 [90% CI]	p_{TOST}
TS-JEPA (Unconstrained)	0.206 ± 0.025	0.331	0.093	0.477	Baseline	–
Operator-Entropy JEPA	0.233 ± 0.028	0.351	0.101	0.550	-0.004 $[-.011, +.003]$	0.0007^{*}
$\lambda_{\text{ent}} = 0$ control	0.235 ± 0.038	0.365	0.099	0.580	–	$0.0088^{\dagger}$
Reynolds-Stress JEPA	0.148 ± 0.022	0.285	0.190	0.421	-0.009 $[-.052, +.032]$	$0.211^{\ddagger}$
$\alpha_{\text{stress}} = 0$ control	0.167 ± 0.012	0.281	0.109	0.419	–	–
TimesNet (Baseline)	0.197 ± 0.018	0.329	0.300	0.583	–	–
TranAD (Baseline)	0.130 ± 0.004	0.287	0.307	0.490	–	–

* TOST equiv. at $\Delta = \pm\ 0.03$ ($p<0.001$); † to ± 0.010 ($p=0.0088$); ‡ Wilcoxon $p=1.00$ (3 seeds).

(b) Dyadic-Shell Front-End On/Off

Stream (F1 / PR-AUC)	Std. Conv	Dyadic Shells	ΔF1 / ΔPR
SMAP `P-3`	**0.301** / 0.280	0.115 / **0.296**	-0.186 / $+0.017$
MSL `M-1`	0.014 / **0.491**	**0.016** / 0.449	$+0.002$ / -0.043
SMD `machine-1-2`	0.000 / 0.028	**0.068** / **0.221**	$+0.068$ / $+0.193$
Daphnet `S01R01E1`	**0.179** / **0.103**	0.130 / 0.066	-0.048 / -0.037
GECCO `water_quality`	**0.097** / 0.161	0.062 / **0.282**	-0.035 / $+0.121$
Macro Mean	**0.118** / 0.213	0.078 / **0.263**	-0.040 / $+0.050$

(c) Factorial and Strength Sweep of the Operator Regularizer

			Operator / Representation / Stability				Buffered			Causal		
Arm	λ_{ent}	β_{norm}	RankMe($\mathbf{K}$)	$\kappa(\mathbf{K})$	RankMe($z_{\text{ctx}}/z_{\text{tgt}}$)	$\text{CV}_{\mathcal{L}}$	FPR	F1	ER	FPR	F1	ER
Neither	0	0	31.53	1.437	5.16/11.57	0.565	0.071	0.044	0.250	0.017	0.009	0.369
Entropy only	0.5	0	31.84	1.256	5.25/11.57	0.550	0.071	0.040	0.266	0.017	0.011	0.389
Barrier only	0	1	31.54	1.431	5.37/11.58	0.563	0.074	0.047	0.266	0.016	0.011	0.389
Both	0.05	1	31.61	1.395	5.16/11.57	0.562	0.074	0.039	0.250	0.017	0.008	0.373
Both	0.10	1	31.67	1.363	5.19/11.58	0.559	0.073	0.039	0.250	0.017	0.008	0.421
Both (deployed)	0.50	1	31.85	1.255	5.54/11.56	0.549	0.072	0.038	0.266	0.017	0.010	0.444
Both	1.00	1	31.91	1.204	5.25/11.55	0.546	0.070	0.037	0.250	0.017	0.007	0.365
Both	2.00	1	31.95	1.154	5.34/11.54	0.545	0.072	0.032	0.250	0.017	0.004	0.397

(d) Quiet-Fault Point Recall

Injected fault	Unif.	Var.	MAD	`lowvar`	`dev_maxz`
Flatline	0.021	0.043	0.058	0.182	0.087
Flatline (mid-var.)	0.024	0.048	0.068	0.317	0.087
Stuck sensor	0.021	0.043	0.058	0.182	0.087
Dropout	0.020	0.043	0.058	0.182	0.087
Low-amp. step	0.030	0.049	0.062	0.182	0.087
$+5\sigma$ spike	0.023	0.042	0.059	0.182	0.108
Mean nom. FPR	0.019	0.069	0.055	0.149	0.141

(e) Controlled Mechanism-Discrimination Study (19 Non-GHL Streams)

	Tail/Drift		Held FPR		Point-F1		PR-AUC		Causal Event Recall			
Arm	γ_2	D_{KS}	Caus.	Buf.	Caus.	Buf.	Caus.	Buf.	SPOT	q^{*}=0.05	f^{*}=0.01	f^{*}=0.05
`TS-JEPA`	1.33	0.214	0.024	0.093	0.161	0.206	0.278	0.331	0.487	0.730	0.395	0.584
`OpEntropy`	0.94	0.209	0.028	0.101	0.153	0.233	0.280	**0.351**	0.517	0.752	0.371	0.573
`Reynolds`	2.04	0.274	0.086	0.190	0.060	0.148	0.231	0.285	0.474	0.667	0.281	0.494
`Reynolds (α=0)`	2.41	0.193	0.019	0.109	0.051	0.167	0.220	0.281	0.424	0.756	**0.457**	**0.633**
`JEPA-CtxRecon`	2.59	0.203	0.017	0.108	0.112	0.188	0.235	0.256	0.542	0.795	0.431	0.573
`JEPA-NoEMA`	1.16	**0.167**	**0.009**	0.098	0.146	**0.235**	0.273	0.328	0.503	**0.816**	0.398	0.600
`TS-JEPA-NoGate`	0.91	0.214	0.018	0.077	0.156	0.209	0.272	0.324	0.538	0.728	0.418	0.599
`PCA-LinPred-LN`	0.95	0.233	0.021	0.186	0.017	0.104	0.210	0.244	0.452	0.665	0.377	0.624
`TCN-ObsPred-LN`	0.91	0.376	0.117	0.193	0.084	0.126	0.235	0.257	0.421	0.665	0.329	0.541
`IsoForest`	**0.70**	0.373	0.039	**0.056**	0.041	0.041	0.209	0.222	0.149	0.631	0.317	0.443
`TCN-ObsPred`	6.10	0.463	0.247	0.313	0.140	0.145	0.272	0.310	0.546	0.629	0.171	0.383
`TCN-ObsRecon`	6.24	0.445	0.245	0.304	0.134	0.135	0.261	0.301	0.528	0.619	0.163	0.379
`TimesNet`	5.54	0.438	0.243	0.300	**0.209**	0.197	**0.293**	0.329	**0.588**	0.618	0.217	0.407
`TranAD`	5.64	0.459	0.222	0.307	0.140	0.130	0.248	0.287	0.533	0.602	0.149	0.343

4) **Context autoencoding (`JEPA-CtxRecon`) and linear controls:** Autoencoding context embeddings without temporal foresight achieves comparable drift reduction ($D_{\text{KS}} = 0.203$, causal FPR 1.7%, event recall 0.542), while linear projection (`PCA-LinPred-LN`) reaches 2.1% causal FPR at lower F1 (0.017) and `IsoForest` collapses in recall (0.149). LayerNorm geometry thus bounds tail kurtosis, while learned latent projection is required to suppress temporal distribution drift.

VII. RESULTS AND DISCUSSION

A. Benchmark Performance on Continuous Telemetry

Table III reports all 33 continuous telemetry streams under the chronological held-out SPOT protocol ($u = p_{98}, q = 10^{-3}$): Panel (a) uses retrospective buffered attribution, Panel (b) strictly causal streaming, and Panel (c) per-dataset breakdowns. The main result is a large relative drop in false alarms. Point-F1 and PR-AUC against reconstruction baselines are statistically inconclusive.

Fewer False Alarms on Held-Out Telemetry: On the 19 non-GHL streams (Table III(a),(b)), latent scoring lowers held-out nominal FPR three- to four-fold under buffered attribution (9.3% and 10.1% for TS-JEPA and OpEntropy JEPA versus 30.0% and 30.7% for TimesNet and TranAD) and eight- to ten-fold under causal streaming (2.4% and 2.8% versus 24.3% and 22.2%). The reduction holds across all 33 streams in paired tests (buffered FPR -14.4% to -15.3% versus TimesNet, $p \leq 0.006$; 95% CI for OpEntropy $[-24.1\%, -6.5\%]$) and across the $N = 7$ dataset families ($p = 0.016$–0.031). Leaving one dataset out at a time keeps buffered FPR at 3.8%–7.0% for JEPAs versus 8.4%–18.3% for reconstructors. The exceptions are MSL, where TimesNet has the lower buffered FPR (13.3% vs. 17.7%), and Genesis, Metro, and SWAN, where reconstructors emit no alarms at all because their calibration thresholds are too high.

Detection Quality Is Comparable, Not Better: Lower false-alarm rates do not make latent scoring a better detector. Buffered Point-F1 and event recall for TS-JEPA and OpEntropy JEPA fall in the same range as TimesNet and TranAD (Table III(a)), and paired differences across all 33 streams are indistinguishable from zero (Point-F1 $+0.012$ and $+0.051$, $p = 0.54$ and 0.88; PR-AUC -0.000 and $+0.023$, $p = 0.39$ and 0.16 versus TimesNet and TranAD). Per-dataset standard deviations are as large as the means (Table III(c)). Only SMAP beats the trivial always-alarm F1 of 0.236 (0.379 and 0.396), although PR-AUC exceeds base prevalence π on both SMAP (0.504–0.525 vs. $\pi = 0.150$) and MSL (0.422–0.434 vs. $\pi = 0.255$).

Domain and Operational Applicability Boundaries: Synthesizing performance across fault classes and sampling rates defines four operational regimes:

1) *Favorable operating regime:* Multi-channel spacecraft and

TABLE III
HELD-OUT CALIBRATION BENCHMARK (SPOT, $u{=}p_{98}$, $q{=}10^{-3}$): (A) RETROSPECTIVE BUFFERED ($S{=}64$) AND (B) STRICTLY CAUSAL TRAILING ATTRIBUTION ACROSS 19 NON-GHL STREAMS AND 14 GHL STREAMS, (C) PER-DATASET BREAKDOWN ACROSS $N{=}7$ TELEMETRY FAMILIES, AND (D) 130-STREAM SINGLE-SEED CROSS-CHECK. BEST IN **BOLD**, SECOND UNDERLINED.

	(a) Retrospective Buffered, Non-GHL (19 Streams)					(b) Strictly Causal Trailing, Non-GHL (19 Streams)				GHL Buffered (14)		Macro
Architecture	**FPR ↓**	**Point-F1 ↑**	**PR-AUC ↑**	**ER ↑**	**Δt ↓**	**FPR ↓**	**Point-F1 ↑**	**ER ↑**	**Δt ↓**	**FPR ↓**	**Point-F1 ↑**	**DS-F1 ↑**
Latent-space predictive architectures (JEPAs)												
TS-JEPA (unconstrained)	0.093	0.206	0.331	0.477	110.3	0.024	0.161	0.487	124.0	0.004	0.000	0.115
Operator-Entropy JEPA	0.101	**0.233**	**0.351**	0.550	79.5	0.028	0.153	0.517	112.5	0.012	0.001	**0.125**
Reynolds-Stress JEPA	0.190	0.148	0.285	0.421	116.5	0.086	0.060	0.474	133.9	0.009	0.002	0.073
Reynolds ($\alpha_{\text{stress}}{=}0$)	0.109	0.167	0.281	0.419	120.7	**0.019**	0.051	0.424	228.4	–	–	0.121
Observation-space reconstructive baselines												
TimesNet	0.300	0.197	0.329	**0.583**	**48.5**	0.243	**0.209**	**0.588**	**62.2**	0.061	**0.022**	0.100
TranAD	0.307	0.130	0.287	0.490	234.4	0.222	0.140	0.533	241.0	0.046	0.020	0.069
Non-neural linear baseline												
Linear AR Forecaster	**0.005**	0.051	0.255	0.432	87.8	–	–	–	–	**0.002**	0.000	0.038

	(c) Telemetry Family Descriptors				Buffered Point-F1 (Stream Mean) ↑					Buffered Held-Out Nominal FPR ↓				
Dataset	n	K	π	$\text{F1}_{\text{all-pos}}$	**TS-JEPA**	**OpEnt**	**Reynolds**	**TimesNet**	**TranAD**	**TS-JEPA**	**OpEnt**	**Reynolds**	**TimesNet**	**TranAD**
Daphnet	6	9	0.148	0.256	0.145	0.122	0.156	**0.229**	0.144	0.091	0.059	**0.039**	0.313	0.298
GHL	14	16	0.013	0.026	0.000	0.001	0.002	**0.022**	0.020	**0.004**	0.012	0.009	0.061	0.046
Genesis	1	18	0.003	0.006	0.000	**0.008**	0.000	0.000	0.000	0.017	0.082	0.016	**0.000**	**0.000**
MSL	4	55	0.255	0.381	0.161	**0.323**	0.114	0.158	0.151	**0.095**	0.177	0.106	0.133	0.241
SMAP	6	25	0.150	0.236	0.379	**0.396**	0.237	0.289	0.167	0.135	**0.125**	0.486	0.549	0.513
Metro	1	5	0.001	0.003	0.000	0.000	0.000	**0.001**	0.000	0.014	0.026	0.017	**0.000**	**0.000**
SWAN	1	38	0.279	0.437	**0.119**	0.024	0.000	0.000	0.000	**0.000**	**0.000**	**0.000**	**0.000**	**0.000**

Note: 95% stream-bootstrap CIs on 19 non-GHL streams for Panel (a) Buffered FPR / Point-F1: TS-JEPA [.048, .145] / [.113, .319], OpEntropy [.043, .180] / [.131, .355], Reynolds [.092, .304] / [.069, .239], Reynolds ($\alpha_{\text{stress}}{=}0$) [.058, .166] / [.077, .276], TimesNet [.162, .446] / [.098, .315], TranAD [.177, .445] / [.053, .232], Linear AR [.001, .011] / [.012, .104]; for Panel (b) Causal FPR / Point-F1: TS-JEPA [.007, .046] / [.063, .287], OpEntropy [.007, .063] / [.070, .247], Reynolds [.035, .141] / [.028, .100], Reynolds ($\alpha_{\text{stress}}{=}0$) [.006, .038] / [.020, .094], TimesNet [.123, .375] / [.109, .326], TranAD [.119, .337] / [.058, .241]. In Panel (c), within-dataset F1 SDs are 0.058–0.232 (Daphnet), 0.119–0.263 (MSL), and 0.254–0.361 (SMAP). **(d) 130-Stream Single-Seed Cross-Check (Point-F1, dataset-macro / stream mean across 11 datasets):** NCAD 0.058/0.088; DCdetector 0.070/0.097; Anomaly Transformer 0.091/0.082; TimesNet 0.129/0.151; TranAD 0.156/0.152; TS-JEPA **0.160 / 0.186**.

wearable kinematic arrays ($K \geq 9$, SMAP, MSL, Daphnet) with persistent localized transients, where suppressing nuisance false alarms under automated calibration is the primary operational requirement.

2) *Latency-constrained telemetry:* Systems where a 63-step emission lag is prohibitive for immediate tripwire response, requiring short horizons $S \in \{1, 4\}$ (Section VII-D).
3) *Quiet single-channel faults:* Inactive sensor modes (flatlines, dropouts) that LayerNorm embeddings absorb as nominal background, requiring auxiliary variance monitors when single-channel coverage is mandatory (Section VI-C).
4) *Continuous non-stationary process ramps:* Industrial heating loops and traffic drifts (GHL, Metro; 0.000–0.022 F1), which defeat both static and rolling thresholds.

B. Evaluating Physics-Inspired Structures: Operator Conditioning and Reynolds Closure

What do the Operator-Entropy and Reynolds-Stress formulations contribute beyond unconstrained `TS-JEPA`? As established by the fixed-backbone equivalence tests in Section VI-A, the spectral regularizer conditions the Koopman matrix $\mathbf{K}$ without altering detection accuracy ($p_{\text{TOST}} < 0.01$ down to $\Delta = \pm 0.010$). Similarly, on the Reynolds architecture, isolating the base prediction residual ($\alpha_{\text{stress}} = 0$) yields buffered PR-AUC of 0.281 versus 0.285 and Point-F1 of 0.167 versus 0.148 (paired difference -0.009, 90% CI $[-0.052, +0.032]$) while lowering causal held-out FPR from 0.086 to 0.019 (Table III(a),(b)). Both models provide concrete structural properties: OpEntropy JEPA shortens mean buffered detection delay from 110.3 to 79.5 steps (124.0 → 112.5 causally, Table III(b), and 127.0 → 58.2 on the decoupling panel), while Reynolds-Stress JEPA structurally bounds the numerical range inside the Crouzeix contractive disk and monitors second-moment covariance departures. These physics-inspired structures act as principled geometric inductive biases rather than raw accuracy drivers.

C. Operational Incident Evaluation and Causal-vs-Buffered Trade-offs

Comparing Panels (a) and (b) of Table III isolates the operational cost of zero-lookahead streaming. Without retrospective smoothing, the lower false-alarm rate (2.4%–2.8%) trades onset speed and sensitivity: causal Point-F1 and event recall are more conservative (0.153–0.161 and 0.487–0.517, vs. 0.209 and 0.588 for TimesNet), and mean detection delay increases to 112.5–124.0 steps (62.2 for TimesNet). Latent scoring thus functions in streaming mode as a conservative filter that prioritizes false-alarm suppression over rapid onset alarm. On five non-stationary incident streams, pairing latent scores with dynamic rolling SPOT suppresses false alarms by up to 21× under regime drift.

D. Horizon Length, Score-Emission Latency, and False-Alarm Control

As established in Section IV-C, target horizon $S = 64$ imposes a 63-step emission lag before an alarm can trigger. To evaluate whether shorter horizons eliminate this latency without sacrificing false-alarm suppression, Fig. 3(a) retrains models across $S \in \{1, 4, 16, 32, 64\}$ on five streams (three seeds). At $S = 1$ (zero emission lag), OpEntropy JEPA holds causal FPR to 0.5% (vs. 2.5% for TimesNet) while achieving 0.729 event recall and a detection delay of $\Delta t = 76.3$

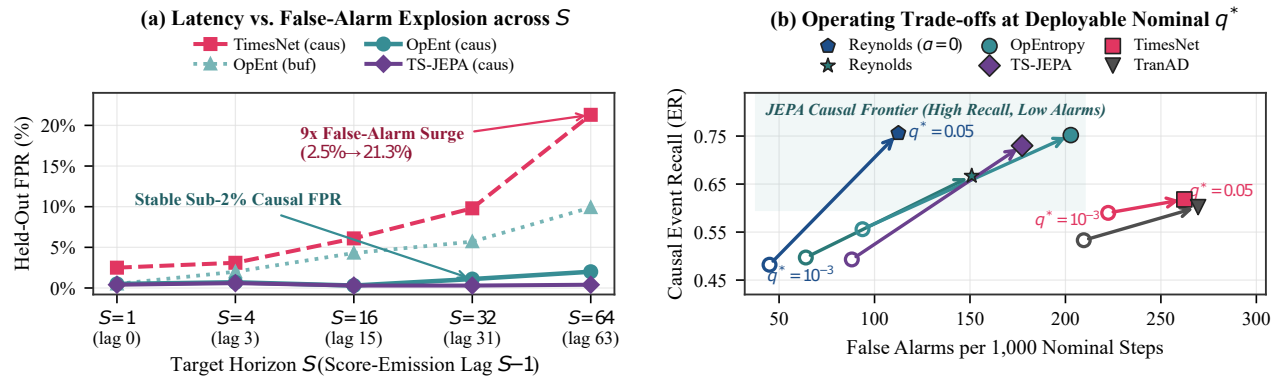


Fig. 3. Operational latency, score emission, and deployable operating trade-offs. **(a)** Held-out false-positive rate across target horizon lengths $S \in \{1, 4, 16, 32, 64\}$ (score-emission lag $S-1 \in \{0, 3, 15, 31, 63\}$ steps, five streams): TimesNet false alarms explode nine-fold (2.5% → 21.3%), while TS-JEPA and OpEntropy JEPA remain stable under 2%. At S=1 (zero emission lag), OpEntropy JEPA cuts detection delay to $\Delta t = 76.3$ steps. **(b)** Deployable operating trade-offs at nominal calibration quantiles $q^* \in \{0.001, 0.050\}$ across 19 non-GHL streams (directed arrows indicate q^*=0.001 → 0.050): latent predictors define a superior Pareto frontier, achieving higher causal event recall (0.667–0.756) at substantially fewer nuisance false alarms (113–203) than TimesNet and TranAD (262–270).

steps. As horizon S increases from 1 to 64, causal FPR for TimesNet rises nearly nine-fold (2.5% → 21.3%), whereas OpEntropy and TS-JEPA remain bounded within 0.3%–2.0%. Time-critical monitoring applications can therefore operate at $S \in \{1, 4\}$ to eliminate score-emission latency while preserving relative false-alarm suppression.

E. Deployable Calibration Quantiles vs. Matched Realized Test-FPR Operating Points

Reconstruction baselines calibrate to a realized false-alarm rate of 22%–31%, against 1%–10% for latent predictors. Raw recall at one SPOT threshold therefore mixes detector quality with threshold placement. Long anomaly segments make this worse: Daphnet segments average 402 steps and SMAP segments 1,102, so a detector alarming at random with FPR 30% still touches over 95% of segments, since $1-(1-\text{FPR})^L$ approaches one. We compare the models under two operating regimes:

1) **Deployable nominal calibration quantiles (q^*, Fig. 3(b)):** Thresholds are set at fixed empirical quantiles of the nominal calibration block without test labels. At causal calibration quantile $q^* = 0.050$, TS-JEPA and OpEntropy JEPA achieve higher event recall (0.730 and 0.752) and higher precision (0.251 and 0.259) at fewer false alarms (177–203 per 1,000 nominal steps) than TimesNet (0.618 recall, 262 alarms) and TranAD (0.602 recall, 270 alarms).
2) **Post-hoc matched realized test-FPR budgets (f^*, Table II(e)):** Oracle thresholds are chosen to equalize realized test nominal FPR across models. At a stringent 1% realized causal false-alarm budget ($f^* = 0.01$), latent models nearly double event recall (0.371–0.431) compared to TimesNet (0.217) and TranAD (0.149). At relaxed budgets ($f^* \geq 0.10$ causal and across buffered mappings), recall curves converge ($p = 0.625$ over $f^* \in [0.10, 0.50]$). Latent scoring thus specifically improves sensitivity in the low-false-alarm operating regime ($f^* \leq 0.01$).

TABLE IV
THRESHOLD CALIBRATOR COMPARISON AT DESIGN RISK q=10^{-3}: (A) STATIC RULES ACROSS 33 STREAMS (CELL-WEIGHTED MEANS, STREAM-MACRO IN PARENTHESES), (B) CONTROLLED-DRIFT STRESS TEST (n=16), AND (C) THREE-STAGE REFITTED PERMUTATION DECOMPOSITION OF EXCESS CAUSAL FPR (n=18).

(a) Model	Calibrator Rule	Causal FPR	Over.	Buffered FPR	F1 (buf)
TS-JEPA	Standard SPOT	0.017 (0.015)	10×	0.064 (0.055)	0.1386
	SPOT + Kurtosis	0.019 (0.016)	10×	0.163 (0.140)	0.1498
	Empirical p99.5	0.025 (0.023)	21×	0.076 (0.066)	0.1399
	Validation Max	0.016 (0.014)	7×	0.066 (0.057)	0.1385
OpEntropy	Standard SPOT	0.021 (0.020)	14×	0.073 (0.064)	0.1573
	SPOT + Kurtosis	0.021 (0.019)	10×	0.135 (0.117)	0.1610
	Empirical p99.5	0.030 (0.028)	27×	0.088 (0.078)	0.1633
	Validation Max	0.019 (0.017)	10×	0.073 (0.064)	0.1566
Reynolds	Standard SPOT	0.060 (0.052)	52×	0.131 (0.113)	0.1003
	SPOT + Kurtosis	0.056 (0.048)	48×	0.179 (0.153)	0.0987
	Empirical p99.5	0.069 (0.061)	61×	0.143 (0.125)	0.1027
	Validation Max	0.060 (0.052)	52×	0.132 (0.114)	0.0968
TimesNet	Standard SPOT	0.180 (0.161)	49×	0.222 (0.199)	0.1395
	SPOT + Kurtosis	0.184 (0.162)	46×	0.228 (0.201)	0.1352
	Empirical p99.5	0.191 (0.175)	51×	0.234 (0.213)	0.1413
	Validation Max	0.179 (0.160)	48×	0.226 (0.203)	0.1410
TranAD	Standard SPOT	0.160 (0.142)	46×	0.222 (0.196)	0.0936
	SPOT + Kurtosis	0.172 (0.151)	43×	0.245 (0.215)	0.0940
	Empirical p99.5	0.171 (0.154)	51×	0.234 (0.209)	0.0962
	Validation Max	0.158 (0.139)	44×	0.227 (0.201)	0.0952

(b) Injected condition	Static SPOT	Static emp.	Rolling emp.	Rolling SPOT
Stationary (no injection)	0.061	0.162	0.016	0.047
Mean shift $+1.0\sigma$	0.087	0.216	0.011	0.043
Regime change $+1\sigma$	0.073	0.187	0.015	0.070
Variance scale 2.0	0.082	0.195	0.016	0.047
Autocorrelation 0.9	0.113	0.232	0.024	0.055
Continuous mean ramp	0.075	0.186	0.326	0.385

	Refitted Stage Causal FPR			Share of Excess (%)		
(c) Model	FPR_{chr}	FPR_{blk}	FPR_{pt}	Inter-Blk	Serial	i.i.d.
TS-JEPA	0.0121	0.0058	0.0019	56.6	35.3	8.1
OpEntropy	0.0185	0.0073	0.0022	64.1	29.1	6.9
Reynolds	0.0022	0.0037	0.0030	0.0	25.9	74.1
TimesNet	0.0872	0.0204	0.0037	77.5	19.4	3.1
TranAD	0.0741	0.0067	0.0032	92.3	4.7	3.0

95% bootstrap CIs for $\text{FPR}_{\text{chr}} - \text{FPR}_{\text{blk}}$: TS-JEPA $+.0063$ $[-.005, +.019]$, OpEnt $+.0112$ $[-.002, +.034]$, Reynolds $-.0015$ $[-.005, +.002]$, TimesNet $+.0668$ $[+.031, +.108]$, TranAD $+.0674$ $[+.035, +.106]$.

F. Automated Threshold Calibration Realities on Non-Stationary Telemetry

Does static extreme-value thresholding achieve its nominal design risk $q = 10^{-3}$ (0.1%) out of sample? It does not. Across all 33 streams (Table IV(a)), standard SPOT achieves stream-macro causal FPR of 1.51% for TS-JEPA and 1.95% for OpEntropy (15–20× the design target), and buffered FPR of 5.54% and 6.38% (55–64×). These rates are an order of magnitude below TimesNet (16.11% causal, 19.85%

buffered) and TranAD (14.23% causal, 19.63% buffered), yet a 1.5%–6.4% false-alarm rate still falls short of strict 0.1% targets. Closing this gap requires pairing latent scores with: (i) block-declustered GPD fitting via the Ferro–Segers extremal index $\hat{\theta} \in (0, 1]$ [30], (ii) online conformal quantile updates under regime shift (Table IV(b)), or (iii) state-conditioned threshold tables. Three diagnostics in Table IV explain the overshoot:

1) **GPD tail stability and quantile/length sensitivity:** On the 19 non-GHL streams at $u = p_{98}$, median calibration exceedance count is $N_u = 28$ causally (21 buffered) with 0% numerical solver fallback and low bootstrap threshold uncertainty ($\mathrm{SE}(\tau)/\tau = 2.6\%$–$3.5\%$ causal). Causally, Grimshaw GPD fits hit the lower shape boundary ($\xi = -0.90$) in only 4.4% of JEPA cells (median $\xi = -0.27$) versus 34.4% for TimesNet and 54.4% for TranAD. Varying $u \in \{p_{95}, p_{98}, p_{99}\}$ leaves causal FPR unchanged (2.41%–2.42% for TS-JEPA), whereas halving calibration length (50% N_{cal}) raises causal FPR to 3.72% (TS-JEPA) and 4.13% (OpEntropy).
2) **Platykurtic collapse under total-threshold kurtosis scaling:** Total-threshold kurtosis scaling $\tau_{\text{eff}} = \tau[1 + \frac{1}{6}\tanh(\gamma_2)]$ reduces buffered FPR on leptokurtic cells ($\gamma_2 \geq 0$, 4.49% $\rightarrow$ 1.11% for TS-JEPA), but 64-step buffered smoothing renders 31.9% of TS-JEPA cells platykurtic ($\gamma_2 < 0$) with tight relative dispersion ($\sigma_{\text{cal}}/\mu_{\text{cal}} \approx 0.02$–$0.05$). Scaling total τ toward zero pulls τ_{eff} below the $u = p_{98}$ floor in 28.4% of cells, inflating overall buffered FPR from 6.40% to 16.30% (Table IV(a)). Scaling strictly the tail excess above u, $\tau_{\text{exc}} = u + (\tau - u)[1 + \frac{1}{6}\tanh(\gamma_2)] \geq u$, prevents threshold collapse below u and holds 19-stream buffered FPR to 9.19% vs. 9.29% unadjusted (2.40% vs. 2.42% causal).
3) **Three-stage refitted permutation decomposition (Table IV(c)):** Simply permuting held-out evaluation scores against a fixed threshold cannot change the exceedance rate. Instead, our three-stage protocol pools the nominal calibration and evaluation vectors $\mathbf{v}_{\text{pool}} = [v_{\text{cal}}, v_{\text{hold}}] \in \mathbb{R}^{N_{\text{cal}}+N_{\text{hold}}}$ and *refits* the SPOT threshold ($u = p_{98}, q = 10^{-3}$) from scratch on each permuted calibration split: Stage 1 (chronological) fits τ_{chrono} on v_{cal} and evaluates $\text{FPR}_{\text{chrono}}$ on v_{hold}; Stage 2 ($L = 64$ block permutation) shuffles contiguous 64-step blocks across $\mathbf{v}_{\text{pool}}$, splits into permuted calibration (N_{cal}) and evaluation (N_{hold}) halves, *refits* τ_{block}, and evaluates $\text{FPR}_{\text{block}}$ (equalizing macro-block distributions while preserving intra-block autocorrelation); and Stage 3 (pointwise i.i.d. permutation) uniformly shuffles all points in $\mathbf{v}_{\text{pool}}$, splits into calibration and evaluation halves, *refits* τ_{point}, and evaluates $\text{FPR}_{\text{point}}$. Across the 18 streams with $N_{\text{cal}}, N_{\text{hold}} \geq 100$ (Table IV(c)), inter-block distribution shift ($\text{FPR}_{\text{chrono}} - \text{FPR}_{\text{block}}$) accounts for an estimated 56.6%–64.1% of excess causal overshoot for JEPAs (77.5%–92.3% for reconstructors), and serial dependence ($\text{FPR}_{\text{block}} - \text{FPR}_{\text{point}}$) for 29.1%–35.3%. Because latent predictors achieve $\text{FPR}_{\text{chrono}} = 0$ on 11 of 18 streams (concentrating stream-level variance onto the seven drifting streams so the TS-JEPA and OpEntropy inter-block 95% CIs in Table IV(c) span zero), these percentages represent estimated shares of aggregate excess risk rather than invariant per-stream causal proportions.

VIII. Conclusion and Future Work

We evaluated how Joint-Embedding Predictive Architectures condition anomaly residuals under automated extreme-value calibration across 33 continuous telemetry streams under a chronological held-out protocol:

1) **Mechanism Isolation and Regularizer Equivalence:** Controlled ablations decouple score compression from representation learning: LayerNorm bottlenecks bound residual kurtosis, whereas learned latent projections halve calibration-to-holdout distribution drift ($D_{\text{KS}} : 0.376 \rightarrow 0.193$–$0.214$ for base latent predictors) and reduce causal false-positive rates to 1.9%–2.8%. On a fixed Koopman backbone, the spectral regularizer satisfies TOST equivalence ($p_{\text{TOST}} < 0.01$ down to $\Delta = \pm 0.010$), while the OpEntropy transition architecture shortens mean detection delay from 110.3 to 79.5 steps ($127.0 \rightarrow 58.2$ causally).
2) **Relative False-Alarm Control vs. Detection Parity:** Latent predictive scoring lowers held-out false alarms three-fold under buffered attribution and eight- to ten-fold under causal streaming, and nearly doubles causal event recall at stringent 1% false-alarm budgets ($f^* = 0.01$), while overall Point-F1 and PR-AUC remain comparable to reconstructive baselines.
3) **Calibration Drift and Operational Boundaries:** Static SPOT still overshoots its nominal 0.1% design risk by 15–64$\times$ out of sample, driven primarily by inter-block distribution shift (56.6%–64.1% of excess risk) and serial dependence (29.1%–35.3%). Future work should pair latent predictive residuals with declustered GPD fitting, online conformal updates, and hybrid variance monitors to close the remaining calibration gap and detect quiet single-channel flatlines.

References

[1] F. Meng, B. Yuan, and F. Gao, "Spacecraft telemetry anomaly detection based on deep learning: A review and benchmark," *IEEE Transactions on Aerospace and Electronic Systems*, vol. 56, no. 5, pp. 3863–3878, 2020.

[2] R. Chalapathy and S. Chawla, "Deep learning for anomaly detection: A survey," *arXiv preprint arXiv:1901.03407*, 2019.

[3] K. Hundman, V. Constantinou, C. Laporte, I. Colwell, and T. Soderstrom, "Detecting spacecraft anomalies using lstms and nonparametric dynamic thresholding," *Proceedings of the 24th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, pp. 387–395, 2018.

[4] J. Xu, H. Wu, J. Wang, and M. Long, "Anomaly transformer: Time series anomaly detection with association discrepancy," in *International Conference on Learning Representations (ICLR)*, 2022.

[5] H. Wu, T. Hu, Y. Liu, H. Zhou, J. Wang, and M. Long, "Timesnet: Temporal 2d-variation modeling for general time series analysis," in *International Conference on Learning Representations (ICLR)*, 2023.

[6] S. Tuli, G. Casale, and N. R. Jennings, "Tranad: Deep transformer networks for anomaly detection in multivariate time series data," *Proceedings of the VLDB Endowment*, vol. 15, no. 6, pp. 1201–1214, 2022.

[7] A. Siffer, P.-A. Fouque, A. Termier, and C. Largouet, "Anomaly detection in streams with extreme value theory," in *Proceedings of the 23rd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 2017, pp. 1067–1075.

[8] Y. LeCun, "A path towards autonomous machine intelligence," *OpenReview Preprint*, 2022.

[9] C. U. Carmona, F.-X. Aubet, V. Flunkert, and J. Gasthaus, "Neural contextual anomaly detection for time series," in *International Conference on Learning Representations (ICLR)*, 2022.
[10] S. Schmidl, P. Wenig, and T. Papenbrock, "Anomaly detection in time series: a comprehensive evaluation," *Proceedings of the VLDB Endowment*, vol. 15, no. 9, pp. 1779–1797, 2022.
[11] Z. Yue, Y. Wang, J. Duan, T. Yang, C. Huang, Y. Tong, and B. Xu, "Ts2vec: Towards universal representation of time series," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 8, 2022, pp. 8980–8987.
[12] Y. Yang, C. Zhang, T. Zhou, Q. Gao, Q. Wen, and Y. Liang, "Dcdetector: Dual attention contrastive representation learning for time series anomaly detection," in *Proceedings of the 29th ACM SIGKDD Conference on Knowledge Discovery and Data Mining*, 2023, pp. 3033–3045.
[13] M. Assran, Q. Duval, I. Misra, P. Bojanowski, P. Vincent, M. Rabbat, Y. LeCun, and N. Ballas, "Self-supervised learning from images with a joint-embedding predictive architecture," in *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023, pp. 15 619–15 629.
[14] A. Bardes, Q. Duval, C. Feichtenhofer, I. Laptev, D. Haziza, V. Khalidov, N. Ballas, and Y. LeCun, "V-jepa: Video joint-embedding predictive architecture," *arXiv preprint arXiv:2404.08471*, 2024.
[15] J.-B. Grill, F. Strub, F. Altché, C. Tallec, P. Richemond, E. Buchatskaya, C. Doersch, B. Avila Pires, Z. Guo, M. Gheshlaghi Azar *et al.*, "Bootstrap your own latent-a new approach to self-supervised learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 33, 2020, pp. 21 271–21 284.
[16] A. Bardes, J. Ponce, and Y. LeCun, "Vicreg: Variance-invariance-covariance regularization for self-supervised learning," in *International Conference on Learning Representations (ICLR)*, 2022.
[17] J. Zbontar, L. Jing, I. Misra, Y. LeCun, and S. Deny, "Barlow twins: Self-supervised learning via redundancy reduction," in *International Conference on Machine Learning (ICML)*, 2021, pp. 12 310–12 320.
[18] A. Ermolov, A. Siarohin, E. Sangineto, and N. Sebe, "Whitening for self-supervised representation learning," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 34, 2021, pp. 8315–8327.
[19] S. L. Brunton, J. L. Proctor, and J. N. Kutz, "Discovering governing equations from data by sparse identification of nonlinear dynamical systems," *Proceedings of the National Academy of Sciences*, vol. 113, no. 15, pp. 3932–3937, 2016.
[20] R. T. Chen, Y. Rubanova, J. Bettencourt, and D. K. Duvenaud, "Neural ordinary differential equations," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 31, 2018, pp. 6571–6583.
[21] M. Raissi, P. Perdikaris, and G. E. Karniadakis, "Physics-informed neural networks: A deep learning framework for solving forward and inverse problems involving nonlinear partial differential equations," *Journal of Computational Physics*, vol. 378, pp. 686–707, 2019.
[22] S. Greydanus, M. Dzamba, and J. Yosinski, "Hamiltonian neural networks," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 32, 2019, pp. 15 379–15 389.
[23] M. Cranmer, S. Greydanus, S. Hoyer, P. Battaglia, D. Spergel, and S. Ho, "Lagrangian neural networks," *arXiv preprint arXiv:2003.04630*, 2020.
[24] S. Wang, Y. Teng, and P. Perdikaris, "Understanding and mitigating gradient pathologies in physics-informed neural networks," *SIAM Journal on Scientific Computing*, vol. 43, no. 5, pp. A3055–A3081, 2021.
[25] B. O. Koopman, "Hamiltonian systems and transformations in Hilbert space," *Proceedings of the National Academy of Sciences*, vol. 17, no. 5, pp. 315–318, 1931.
[26] I. Mezić, "Spectral properties of dynamical systems, model reduction and decompositions," *Nonlinear Dynamics*, vol. 41, no. 1–3, pp. 309–325, 2005.
[27] S. L. Brunton, B. W. Brunton, J. L. Proctor, and J. N. Kutz, "Koopman invariant subspaces and finite-dimensional approximations of nonlinear dynamical systems," *Journal of Nonlinear Science*, vol. 26, pp. 1131–1162, 2016.
[28] B. Lusch, J. N. Kutz, and S. L. Brunton, "Deep learning for universal linear embeddings of nonlinear dynamics," *Nature Communications*, vol. 9, no. 1, p. 4950, 2018.
[29] Y. Liu, C. Li, J. Wang, and M. Long, "Koopa: Learning non-stationary time series dynamics with Koopman predictors," in *Advances in Neural Information Processing Systems (NeurIPS)*, vol. 36, 2023, pp. 13 480–13 492.
[30] C. A. Ferro and J. Segers, "Inference for clusters of extreme values," *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, vol. 65, no. 2, pp. 545–556, 2003.
[31] S. Kim, K. Choi, J. Choi, B. Lee, and S. Yoon, "Towards a rigorous evaluation of time-series anomaly detection," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 36, no. 7, 2022, pp. 7194–7201.
[32] R. Wu and E. J. Keogh, "Current time series anomaly detection benchmarks are flawed and are creating the illusion of progress," *IEEE Transactions on Knowledge and Data Engineering*, vol. 35, no. 3, pp. 2421–2429, 2023.
[33] M. Bächlin, M. Plotnik, D. Roggen, I. Maidan, J. M. Hausdorff, N. Giladi, and G. Tröster, "Wearable assistant for parkinson's disease patients with the freezing of gait symptom," *IEEE Transactions on Information Technology in Biomedicine*, vol. 14, no. 2, pp. 436–446, 2010.
[34] A. Asuncion and D. Newman, "UCI Machine Learning Repository: CalIt2 Building People Counts Data Set," https://archive.ics.uci.edu/dataset/179/calit2+building+people+counts, 2007.
[35] Y. Zhou, C. Brif, and I. Lourentzou, "mTSBench: A Comprehensive Benchmark for Multi-Domain Time Series Anomaly Detection," *Transactions on Machine Learning Research (TMLR)*, 2026, arXiv:2506.21550.
[36] R. A. Angryk, P. C. Martens, B. Aydin, D. Kempton, S. S. Mahajan, S. Basodi, A. Ahmadzadeh, S. Boubrahimi, S. M. Hamdi, M. A. Schuh *et al.*, "Multivariate time series dataset for space weather data analytics," *Scientific Data*, vol. 7, no. 1, p. 227, 2020.
[37] L. M. Candanedo and V. Feldheim, "Accurate occupancy detection of an office room from light, temperature, humidity and co2 measurements using statistical learning models," *Energy and Buildings*, vol. 112, pp. 28–39, 2016.
[38] S. D. Grimshaw, "Computing maximum likelihood estimates for the generalized pareto distribution," *Technometrics*, vol. 35, no. 2, pp. 185–191, 1993.